

Deep Reinforcement Learning for Planner-Free Humanoid Climbing

Soft Actor-Critic for climbing controller without stance planning or motion priors

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Abstract

Physics based controllers can already climb, but each is given part of the answer. The sequence of stances is planned in advance, or the episode starts from a database of hanging poses, or the motion is shaped against captured human climbing. Whether climbing can be learned without any of that has not been reported.

This thesis builds a GPU parallel climbing environment in MJX and trains a humanoid on it with Soft Actor-Critic. The agent is given no stance planner, no pose database, no motion prior and no target position. It has to find the sequence of holds itself, from the reward. The grip is abstracted rather than simulated.

In the final configuration all five runs climbed the route, taking the torso from a spawn height of 1.40 m to (3.84 ± 0.11) m, and every run finished with its hands on the top row of holds. It took forty-three experiments to get there. I describe the reward failure modes I ran into along the way, each with what caused it and what I changed. The contact sequence was discovered rather than supplied, but it does not generalize well.

1 Introduction

1.1 Motivation

Physics based character control has been heavily studied, but most work focuses on walking, running and getting up after a fall. Humanoid climbing enjoyed far less attention, for a few reasons. One hurdle is recording real life climbing. Optical motion capture puts reflective markers on the body and tracks them with cameras placed around the subject. For walking, the subject is in open space and cameras can see them from every angle. On a wall, the climber is pressed against a surface, so the markers on the chest, hips and the front of the legs are hidden, which can result in gaps in the data, at the most important contact points. While walking is only using the ground which is easy to capture, climbing requires a wall with climbing holds or a rock surface, limiting the variations of routes. These challenges limit the amount of real-world climbing data available, which may also help explain why climbing has received less attention in simulation based research.

Even though it is a less heavily studied topic, it poses interesting challenges. Reaching a specific height, or touching the top most climbing hold, is a sparse reward problem, unreachable by random exploration. Rewarding height alone provides a denser signal, but does not solve the exploration problem underneath. grabbing holds, and climbing is extremely hard to stumble upon randomly without additional rewards, which makes reward shaping unavoidable. But shaping for this sparse reward problem, means paying for proximity, which means paying some amount for a stationary actor forever. This makes climbing a task where shaping is essential but very tricky. Another setback is falling. Falling is unavoidable during exploration and if the humanoid falls it has to restart the climb, making it natural to terminate the episode. Even if it is allowed to continue, it decreases the reward by a lot instantly. At the same time, making progress requires letting go of holds and reaching for new ones, which is the action that easily results in falling. This inherent punishment of exploring behavior makes this task hard for an agent to solve, and motivates staying still.

Climbing also has a structure most locomotion tasks do not. Every move commits the body to a particular hold and a particular limb, and that commitment is discrete, a hold is either selected or not. Reaching a hold requires continuous control. The posture the body defines which holds are in reach, and the posture of the body depends on which are grabbed. For example walking on flat ground has available support everywhere, so the choice of foot placement is also continuous.

This thesis started from my research project where building on RoboPianist [1], training a policy controlling a single hand to grab a climbing hold, using the same algorithm. Even though the dexterous hand I used, was not part of this project, it sparked my interest to explore humanoid climbing more. This previous experience provided a foundation for reward shaping, building simulation environments, and solving problems with the learning pipeline of Soft Actor-Critic. Searching for a sparse reward described above means running many experiments, which motivated me to use GPU parallelism to simulate multiple environments, raised challenges in the process as well.

1.2 Problem Statement

Physics based reinforcement learning was used to achieve humanoid climbing in previous works, but those approaches supplied the agent with a part of the task structure, rather the agent discovering it. Previously humanoid climbing was bootstrapped with a planned sequence of stances [2, 3], motion priors and a hangin state data base [4]. The question if climbing is possible without those, remains open. In this thesis I focus on building a learning pipeline where the agent progresses through the climbing route without being given a predefined sequence of holds or poses.

1.3 Research Questions

RQ1. Can humanoid climbing be learned in a single training stage, without a stance planner, a pose dataset or a motion prior?

RQ2. Which reward formulations produce climbing rather than degenerate behaviour, and what failure modes arise?

RQ3. How reliable are single runs in this setting?

1.4 Thesis Outline

Chapter 2 covers the previous climbing controllers and the reward shaping results this work relies on.

Chapter 3 describes the simulation: the humanoid, the wall, the holds and the grip model.

Chapter 4 gives the MDP, the reward and the training setup.

Chapter 5 works through the reward failure modes I ran into, what could have caused them, what I changed, and what the per term logging showed. It also covers two influential environment changes.

Chapter 6 reports the final configuration over five seeds, and tests those policies on routes they were not trained on.

Chapter 7 answers the research questions and states the limitations.

Chapter 8 concludes.

2 Related Work

2.1 Physics-Based Climbing Controllers

Naderi et al. (2017). Naderi et al. [2] achieve humanoid climbing with the use of a graph-based path planner over stances, assignments of holds to limbs. The edges are transitions between them. A sampling-based optimiser then synthesises the physical motion realising each transition. The contact sequence is therefore the output of a search over hold assignments, and the physics problem the controller solves is the execution of a single move between two given stances.

Naderi et al. (2018). A follow-up [3] keeps this hierarchy and makes both levels more efficient by learning models of move-success probability, effortfulness and the low-level control policy, so that the planner can avoid expanding transitions the character is unlikely to complete.

Kang et al. (2025). Kang et al. [4] replace the planner with a trained two-stage pipeline. They first train a hanging policy, which fills a database with correct poses to grasp holds and support the body. The database contains hold positions, postures and grip states. Then the

climbing policy is trained to move the character’s root toward a specified target location, using the hanging-state dataset to initialize episodes.

The key distinction between these approaches is how much of the climbing structure is specified outside the learned controller. Naderi et al. explicitly plan sequences of climbing configurations. Kang et al. remove this high-level route planner but use a hanging state database for initialization, use motion priors for style, and set a specified target. My goal is to remove even more of these supports. The agent is not given a target location or a pre-generated hanging-state dataset, and must discover climbing progress through the task reward and interaction with the environment.

The comparison of the different methods, can be seen in Table 1.

Table 1: Comparison of the task structure provided to physics-based climbing controllers.

	Naderi et al. [2, 3]	Kang et al. [4]
Provided supports	Graph of stances and transitions	Hanging-state dataset, motion prior and target location
What must be learned	Physical motion between prescribed stances	How to move from pre-learned hanging pose toward the target

2.2 Reward Shaping and Reward Hacking

Shaping a sparse objective risks changing the problem being solved.

Potential-based shaping. Ng et al. [5] give the condition under which it does not happen, a term of the form $F(s, a, s') = \gamma\Phi(s') - \Phi(s)$, with γ matching the discount of the underlying MDP, leaves the optimal policy unchanged for any potential Φ . The terms cancel each other out in an episode leaving only the potential at the initial and terminal states, as seen in Equation(1).

$$[H] \sum_{t=0}^{T-1} \gamma^t [\gamma\Phi(s_{t+1}) - \Phi(s_t)] = -\Phi(s_0) + \gamma^T \Phi(s_T). \quad (1)$$

The agent cannot collect anything by moving around in between, so the best policy stays the same. Any Φ works, as long as the term has that shape and γ is the discount the agent is trained with. The guarantee holds only for terms of that form.

Terminal states. Cipollone et al. [6] point out a condition that is easy to miss. The cancelling leaves Φ at the start and at the end. Sometimes the starting point is almost the same in each episode, but the end is not. So if an episode can finish in a state where Φ is not zero, the agent gets extra reward for where it ended, and the guarantee is gone. The usual fix is to set Φ to zero at every terminal state. They also say that keeping the guarantee is not always the goal, and that it can be worth giving up for faster learning.

My reward is a weighted sum of 7 terms, only some of which have the shape Ng et al. require. Section 4.3 gives the terms and Chapter 5 shows what the missing guarantee permits.

RoboPianist. Zakka et al. [1] train a policy controlling two dexterous hands to play the piano in MuJoCo [7]. The task is contact-rich and high dimensional like climbing is, and it is learned from a dense reward with several hand-designed terms to solve their sparse reward problem. The hands have to press the right keys at the right time, but the musical sequence is provided rather than discovered by the policy and the finger-to-note assignments are predetermined.

RoboPianist is also the software framework on which this work is based. The environment structure, Soft Actor-Critic implementation, and associated hyperparameters are inherited from the RoboPianist codebase [1].

2.3 Research Gap

Kang et al. [4] show that hold selection can be learned rather than planned, but do so with two supports. A hanging policy with poses that initialise the climbing policy, and an adversarial motion prior [8] trained on captured human climbing. Whether either is necessary is not established. No result exists for the case where both are removed. This thesis asks whether climbing behaviour can emerge, and examines what reward structure is required without these forms of prior guidance.

3 Simulation Environment

3.1 Overview

Table 2: Key parameters of the simulation, as used for the final configuration (E43).

Group	Parameter	Value
Simulation	physics engine	MuJoCo 3.5 / MJX (JAX)
	solver iterations	10 (line search 6)
	physics timestep	0.003 s
	frame skip	5
	control timestep Δt	0.015 s (67 Hz)
Robot	model	gym Humanoid
	actuated DOF	17
	bodies / n_q / n_v	14 / 24 / 23
	total mass	42.1 kg
	observation / action dim	392 / 17
	control range	± 0.4
	arm torque limit	80 N m
	hip _z /kneetorque limit	120 / 80 N m
Wall	inclination	-15° (slab)
	face offset x	0.5 m
	width $\times$ length	2.0×5.2 m
Holds	grid (rows $\times$ cols)	12×4 (48 holds)
	first / last row, up the face	0.3 m / 4.7 m
	first / last row, vertical	0.29 m / 4.54 m
	column spacing	0.4 m
	radius / furthest point from wall surface	0.05 m / 0.08 m
Contact	grip clearance	0.04 m (surface-to-surface)
	assist stiffness / damping	1200 N/m / 50
	assist force limit	400 N
Episode	length	1000 steps (15 s)
	fall threshold	torso $z < 1.0$ m
	fall penalty	-10
	per-hold reward timeout	4 s
	spawn position (x, y, z)	(0.05, 0, 1.40) m
	reset noise	$\pm 10^{-2}$, uniform on <code>qpos</code> and <code>qvel</code>

In Table 2, I listed the relevant settings of the simulation during training the most successful actor policy. The parameters n_q and n_v are the lengths of MuJoCo’s generalised position and velocity vectors. They differ by one because the free root stores its orientation as a quaternion, which needs four numbers for position and three for velocity. As seen in Figure 1, the MJX simulation spawns the robot standing in front of the wall.

One episode is 1000 steps, 15 seconds. If the torso falls below 1 m the episode ends with -10 penalty.



Figure 1: The humanoid at spawn, standing on the ground in front of the -15° slab.

3.2 Robot Model

The gym humanoid robot was used for the experiments, because its a well studied, widely used standard robot. It has 17 DOF. 3 abdomen motors, 6 hips, 2 knees, 4 shoulders, 2 elbows. The robot has a free root, not actuated. The humanoid has 14 bodies (including world body), torso, lwaist, pelvis, 2 times thigh, 2 times shin, 2 times foot, 2 times upper_arm and lower_arm. A graphic of the different joints connecting the bodies can be seen in Figure 2. The model is the Gymnasium Humanoid environment [9].¹

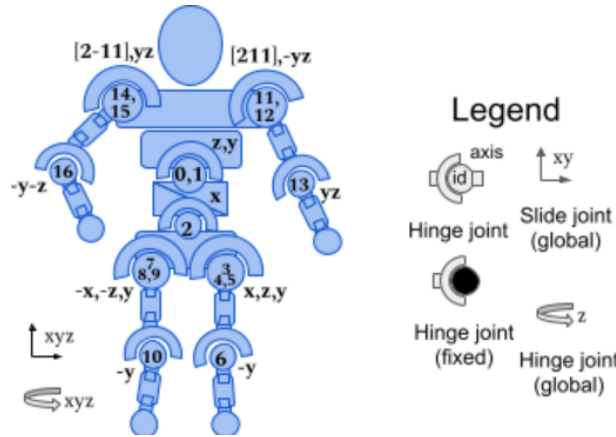


Figure 2: The Gymnasium Humanoid: the 17 actuated joints connecting its 14 bodies [9].

It weights 42.1 kg and has the original 376-dimensional observation of the Gymnasium model [9]. It has a ± 0.4 control range to avoid jittering. The arm actuator torque limits were increased from the original configuration to make the model capable of pulling up its own body weight during climbing. The original torque limits of the gym humanoid, were previously

¹<https://gymnasium.farama.org/environments/mujoco/humanoid/>

optimised for walking, with weak hands. Both the arms and feet have 80 Nm torque limit in the final configuration, and the hip_z has 120N. The assist attaches the limb to the hold. Supporting the body still runs through the arm, but a straight-armed hang is carried largely by the joint limits, so what the arm actuators are needed for is pulling up rather than hanging. I chose a standard simple robot for this task, as previous climbing papers also use simple humanoids [2, 3, 4], that are not designed for sim-to-real transfer.

3.3 Wall and hold representation

I built a similar wall to the one built by Kang et al. [4], shown in Figure 3. I trained with -15 degrees of inclination. I used spherical identical holds that serve as a reward and contact target, evenly spaced on the wall to provide a simple and consistent climbing environment. There is 40 cm of space between rows and columns.

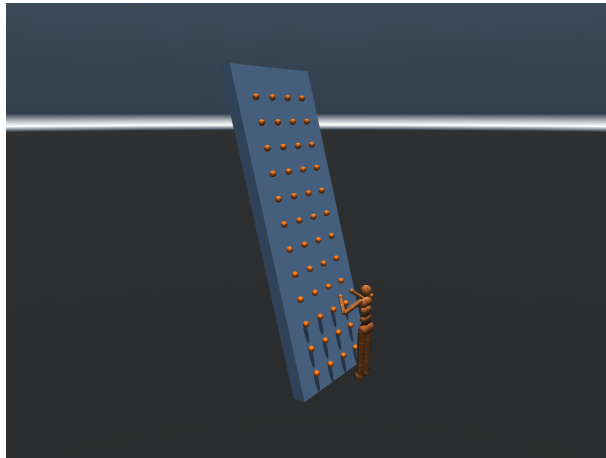


Figure 3: The inclined slab and hold layout, built based on Kang et al. [4].

Each sphere has a 5 cm radius. the highest row of holds is at 4.7 m. The spheres furthest point is 8 cm away from the wall's surface (Figure 4).

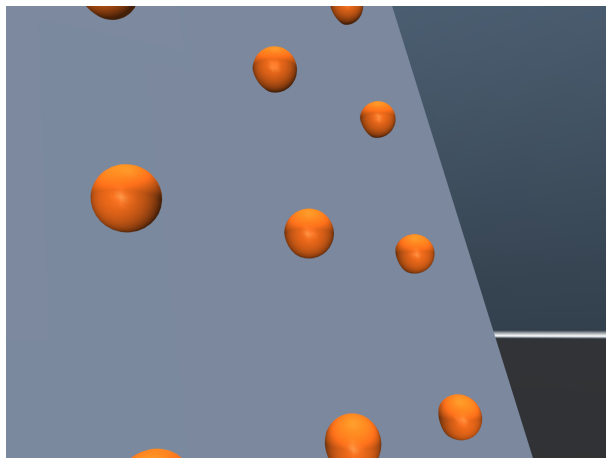


Figure 4: A hold on the wall face. Each hold is a sphere of radius 5 cm whose furthest point sits 8 cm from the wall surface.

As the grip is simulated without fingers and actual grabbing, complex holds are unnecessary, and it would be expensive to use meshes just for visuals, without adding information to the training.

3.4 Grip Model: The Assist Force

In order to make climbing possible the limbs stick to the holds. To make the climbing task less sparse, the holds use an assisting pull to simulate the grab. When the limb is 4 cm from the holds surface, the assist switches on and pulls the limb towards the hold. The assist also simulates the grab, as high friction is not enough for two spheres to simulate it. The assist pulls the limb toward the holds center, following Equation (2), with less force when closer, also resisting the limbs motion, for a smooth grab.

$$\mathbf{F} = g (K_p \mathbf{d} - K_d \mathbf{v}), \quad \|\mathbf{F}\| \leq F_{\max} \quad (2)$$

where $\mathbf{d}$ is the vector from the limb sphere's center to the hold's center, $\mathbf{v}$ is the limb's velocity, and g is 1 if the limb is in the grip radius. $\delta = \|\mathbf{d}\| - r_{\text{limb}} - r_{\text{hold}}$ is the distance of surfaces. The assist engages when $\delta < 4$ cm, but the restoring force is proportional to $\|\mathbf{d}\|$, not to δ . The stiffness is $K_p = 1200 \text{ N m}^{-1}$, the damping $K_d = 50 \text{ N s m}^{-1}$, and the cap $F_{\max} = 400 \text{ N}$. For a limb ($r = 4 \text{ cm}$) on a hold ($r = 5 \text{ cm}$), engagement therefore occurs at $\|\mathbf{d}\| = 0.13 \text{ m}$, giving 156 N, falling to 108 N at surface contact. If the force was proportional to δ , the force would be 0 at contact, but the goal is to stick to the hold, not to get close and fall. The grip assist engages with δ , because limbs are not the same size., and I preferred a uniform engagement across limb sizes. Figure 5 shows the two distances the assist is defined over.

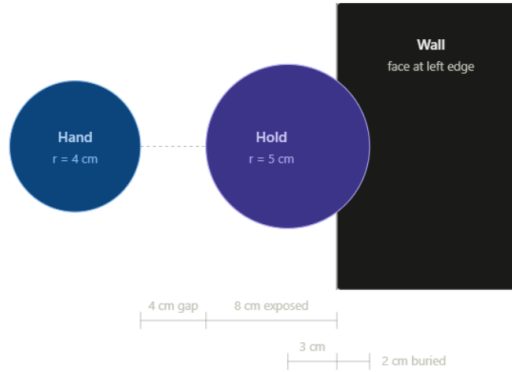


Figure 5: Grip geometry. $\mathbf{d}$ is the centre-to-centre vector and δ the surface-to-surface clearance; the assist engages at $\delta < 4$ cm but the restoring force is proportional to $\|\mathbf{d}\|$.

There is no release action. To let go, the policy has to pull the limb away, until δ exceeds 4 cm, at which point g becomes zero and the assist switches off entirely. That means working against 108 N to 156 N for a hand and 150 N to 198 N for a foot, and the damping term adding $K_d v$ against the motion. The arm actuators can supply 256 N at the hand (Section ??), so a limb can be pulled free, but only deliberately. The 400 N cap therefore never is not used in normal operation, as the spring term alone would need $\|\mathbf{d}\| = 0.33 \text{ m}$, outside the range where the assist acts. It guards against numerical blow-ups rather than limiting the grip.

3.5 Implementation Notes

I used MJX, the JAX re-implementation of MuJoCo [7, 10], for parallel simulation across multiple environments on the GPU. I started with the source code from my research project, which was built using the robopianist source code [1]. As I didn't use GPU for that project, I had to alter the code heavily, and vmap over multiple environments. I used the ITu high performance computing cluster to run the trainings. The simulation step was JIT-compiled using `jax.jit` and vectorized across environments using `jax.vmap`. This allowed multiple environments to be simulated simultaneously while keeping the computation on the GPU, but imposed several constraints that shaped the configuration. The typical solver used for the gymnasium humanoid also needed to be changed because it produces a large compiled program exceeding memory. To reduce the memory overhead, at the cost of a less little accuracy, I used a semi-implicit Euler integrator with a conjugate-gradient (CG) solver limited to 10 iterations and 6 line-search iterations.

4 Method

4.1 Soft-Actor critic

I used the Soft Actor-Critic [11] implementation from RoboPianist [1]. I was familiar with it from the research project, and it had been used successfully there on a contact-rich task with a sparse reward problem.

Soft Actor-Critic is an off-policy actor-critic method for continuous control. It maximises the discounted return with an added entropy term, so the objective is

$$J(\pi) = \mathbb{E}_{\pi} \left[\sum_t \gamma^t (r_t + \alpha \mathcal{H}(\pi(\cdot | s_t))) \right], \quad (3)$$

where $\mathcal{H}$ is the entropy of the policy at state s_t and α controls how much it is worth. A policy that keeps its entropy high keeps trying actions it is not yet sure about, which matters here because most of what the humanoid can do at the start of training ends in a fall.

Being off-policy means the algorithm learns from transitions stored in a replay buffer rather than only from the most recent ones. That decouples the number of gradient updates from the rate at which the simulator produces data, so the two can be set independently. With 512 environments stepping in parallel on a GPU, environment steps are cheap and gradient updates are the bottleneck, so the ratio between them is a parameter that has to be set intentionally (Section 4.4).

The temperature α is not fixed. It is adjusted during training so that the policy's average entropy stays near a target value [12], set here to $-|\mathcal{A}| = -17$. This removes the need to tune α by hand, and it reduces the reward-scale sensitivity of the original formulation [11], in which α acted as an implicit scale on the reward.

Two critics are trained rather than one, and the smaller of their two estimates is used wherever a value is needed. This limits the overestimation that arises when a maximum is taken over noisy estimates [13]. Both critics use layer normalisation.

Each critic is trained by regression against a target formed from the observed reward and the next state,

$$y = r + \gamma(1 - d) \left[\min_{i \in \{1, 2\}} Q_{\bar{\theta}_i}(s', a') - \alpha \log \pi(a' | s') \right], \quad a' \sim \pi(\cdot | s'), \quad (4)$$

where $\bar{\theta}$ denotes the slowly-updated target networks and d indicates termination. The entropy

term appears in the target as well as in the objective, so the critic learns the value of reward and entropy together rather than reward alone.

4.2 MDP Formulation

Table 3: Composition of the 392-dimensional observation. The first six blocks reproduce the Gymnasium Humanoid layout, computed over the 14 humanoid bodies only; the final two are specific to the climbing task.

Block	Source	Dims
Joint positions	<code>qpos[2:]</code> (root x, y omitted)	22
Joint velocities	<code>qvel</code>	23
Body inertias	<code>cinert</code> (14×10)	140
Body velocities	<code>cvel</code> (14×6)	84
Actuator forces	<code>qfrc_actuator</code>	23
External contact forces*	<code>cfrc_ext</code> (14×6)	84
<i>Proprioceptive subtotal</i>		<i>376</i>
Hold-relative vectors	per limb, to nearest active hold (4×3)	12
Grip indicators	per limb, binary	4
Total		392

*MJX does not populate `cfrc_ext`; these 84 values are identically zero. See Section 7.3.

I model climbing as a finite-horizon Markov Decision Process (MDP) defined by the tuple $(\mathcal{S}, \mathcal{A}, \rho, p, r, \gamma, H)$. The policy observes $o \in \mathbb{R}^{392}$ rather than the full simulator state. This consists of the proprioceptive state of the humanoid — joint positions (22), joint velocities (23), body inertias (140), body velocities (84), actuator forces (23) and external contact forces (84); Table 3 gives the full composition. . Episodes run to a maximum of $H = 1000$ steps and may end earlier on a fall. The per-hold payout timers are not part of the observation. They decide which holds still pay, and so determine both the reward and which hold each limb’s vector points at, meaning two observations that look identical can be worth different amounts. I treat the problem as an MDP over the observation, which is standard practice, but it is an approximation.

To make climbing easier, for the end of every limb, I added 3 observations of relative position, to the closest hold that is still active, and pays rewards. (12) This observation helps to learn reaching for a hold, but when the hold stops paying, It needs to differentiate between grabbing a hold and hanging freely. Thats why each limb has a binary observation (4), for grabbing, encouraging the robot to learn letting go. The action space $\mathcal{A} = [-1, 1]^{17}$ represents torque commands for the seventeen actuated joints of the torso, legs and arms, mapped to each actuator’s control range of ± 0.4 and multiplied by that actuator’s gear, giving the applied torque. The state distribution ρ is a fixed standing pose with a uniform noise of $\pm 10^{-2}$ on joint positions and velocities; p is given by the MuJoCo rigid-body dynamics; and r is the reward function. I use a discount factor $\gamma = 0.99$. On termination the transition is treated as terminal and the bootstrap term is dropped, while a time-limit truncation bootstraps normally.

4.3 Reward Formulation

The reward is the weighted sum of seven terms in Equation (5), evaluated at every control step, There is also a -10 penalty at termination for a fall:

$$r_t = \underbrace{10 \phi_{\text{prog}} + 4 \phi_{\text{hands}} + 0.05 \phi_{\text{feet}} + 1.5 \phi_{\text{reach}}}_{\text{potential-based shaping}} + \underbrace{0.5 \phi_{\text{grip}}}_{\text{contact}} + \underbrace{0.02 \phi_{\text{torque}} + 0.01 \phi_{\text{smooth}}}_{\text{effort}} \quad (5)$$

The seven terms are defined in Table 4.

Table 4: Reward terms. Φ denotes a potential; Δt is the control timestep; $w_h \in [1, 2.5]$ is a hold’s height weight, rising linearly from the lowest hold to the highest. The final column indicates whether a term has the form $\Phi(s') - \Phi(s)$, not whether the policy-invariance guarantee of Ng et al. applies to it.

Term	Definition	Weight	Potential-based
Progress	$(\Phi_z(s') - \Phi_z(s))/\Delta t$, $\Phi_z = z_{\text{torso}}$	10.0	yes
Hands	$(\Phi_h(s') - \Phi_h(s))/\Delta t$, $\Phi_h = \bar{z}_{\text{hands}}$	4.0	yes
Feet	$(\Phi_f(s') - \Phi_f(s))/\Delta t$, $\Phi_f = \bar{z}_{\text{feet}}$	0.05	yes
Reach	$(\Phi_r(s') - \Phi_r(s))/\Delta t$, $\Phi_r = \bar{z}_{\text{reach}}$	1.5	yes
Grip	$\sum_i w_i g_i$	0.5	no
Control	$-\ \mathbf{a}\ ^2$	0.02	no
Smoothness	$-\ \mathbf{a}_t - \mathbf{a}_{t-1}\ ^2$	0.01	no
Fall	-10, once, on the terminating step		no

In the grip term, g_i is 1 when the limb is gripping, and 0 when it doesn’t.

Potential-based terms. The first four terms are using $F = \Phi(s') - \Phi(s)$ introduced by Ng et al. [5]. Because consecutive terms cancel, only the net change is paid, so rising up and going down earns nothing. Progress uses the torso height, the hand and foot terms use the mean height the corresponding limbs, so raising a limb toward the next hold pays, while holding it still pays nothing. Division by Δt converts each into a velocity-like quantity, making the shaping term comparable across different control timesteps. The reach term is potential based, it measures the distance from limbs to holds that are still paying, using the potential in Equation (6). A hold stops paying four second after touch.

$$\Phi_r = \frac{1}{4} \sum_{i=1}^4 \sum_{h \in H_{\text{active}}} w_h e^{-d_{ih}/\lambda}, \quad \lambda = 0.5 \text{ m} \quad (6)$$

where d_{ih} is the distance from limb i to the top of hold h . w_h is the wight of the given hold, based on height from 1 to 2.5. λ controls how far a hold’s influence extends. A hold contributes its full weight when a limb touches it. (Section 5).

Non-potential terms. Grip, control, smoothness are not potential based. The grip bonus pays each limb within gripping distance of an active hold, multiplied by that hold’s height so grabbing a higher hold is worth more than grabbing a lower one. $g_i \in \{0, 1\}$, 1 if limb i ’s surface is within 4 cm of a hold’s surface, 0 otherwise. I explain why grip is not a potential based reward term in Section 5.2.

The control term penalizes the size of the control signal. Smoothness penalizes the amount by which the signal changes between consecutive steps. $\mathbf{a}_t \in [-1, 1]^{17}$ is the raw action produced by the policy at step t .

Because there are non potential based reward terms, the final reward is not policy invariant.

4.4 Training Setup

The GPU performance varied greatly, depending on the hardware. I used the ITU high performance cluster, leaving the slurm scheduler to choose the first free computer, as I didn’t want my tasks to queue. Table 5 shows the five runs with the final reward functions. The three hardwares the were running on are NVIDIA A30 NVIDIA A100-PCIE-40GB Quadro RTX 8000.

Table 5: Training hardware and runtime. All runs completed 1.2×10^8 environment steps.

Run	GPU	Hours
E43/1	Quadro RTX 8000	10.1
E43/2	Quadro RTX 8000	10.4
E43/3	NVIDIA A100-PCIE-40GB	7.5
E43/6	NVIDIA A100-PCIE-40GB	7.4
E43/7	NVIDIA A30	9.4
Total		44.8

The Gpu utilization was very high for some machines, and lower for others, making it harder to choose the best environment number (Figure 6).

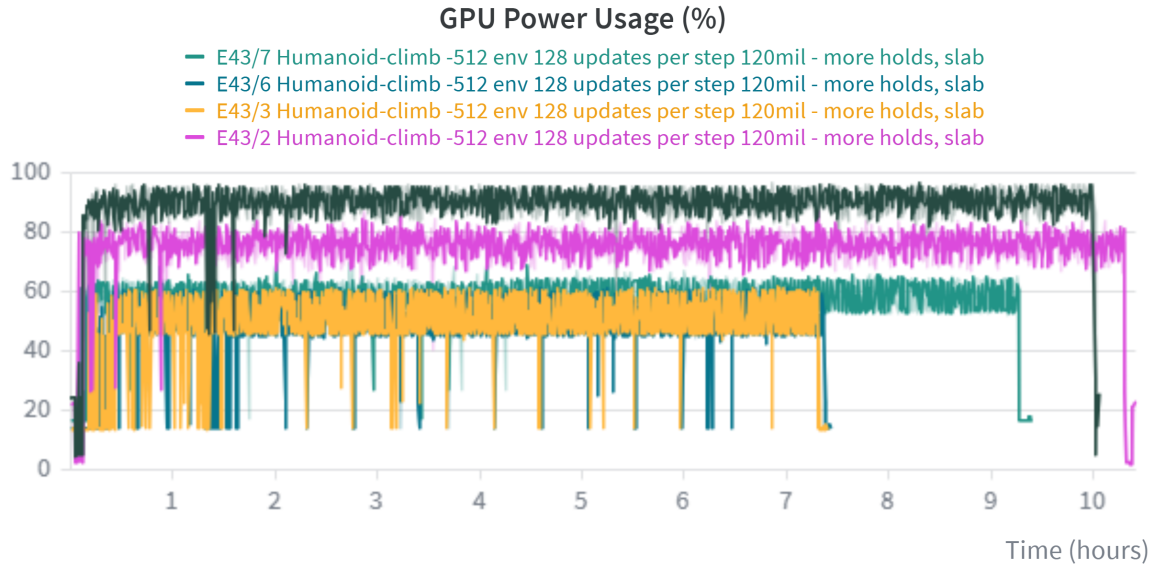


Figure 6: GPU utilisation across the five E43 runs. Utilisation varies with the hardware the scheduler assigned.

The hyperparameters are visible in Table 6.

Table 6: SAC hyperparameters.

Hyperparameter	Value
Total train steps	120M
Random seed	Run specific
Optimizer	
Type	Adam
Learning rate (Actor)	3×10^{-4}
Learning rate (Critic)	3×10^{-4}
Learning rate (Temperature)	3×10^{-4}
β_1	0.9
β_2	0.999
Gradient clipping (global norm)	1.0
Critic	
Hidden units	256
Hidden layers	3
Non-linearity	GELU
Number of critics	2
Layer normalisation	True
Actor	
Hidden units	256
Hidden layers	3
Non-linearity	GELU
Actor log std dev. bounds	$[-20, 2]$
Misc	
Discount factor	0.99
Minibatch size	128
Parallel environments	512
Update-to-data ratio (UTD)	0.25
Replay buffer capacity	1M
Warm-start (seed) steps	100,000
Critic target update frequency	1
Actor update frequency	1
Critic target EMA momentum (τ)	0.005
Target entropy	$- \mathcal{A} $
Entropy temperature (initial)	1.0
Learnable temperature	True
Entropy backup	True

Most of the configuration follows standard practice for SAC on continuous-control tasks and was the same across all runs. Standard SAC performs roughly one gradient update per environment transition. With 512 environments running in parallel, transitions are almost free and gradient updates are the bottleneck, so the two are decoupled. Each rollout collects 16 steps from each of 512 environments, 8192 transitions in all, and 128 gradient updates run against

it, giving 1 875 000 updates over the full budget. That total is ordinary for a SAC run; what differs is how much data those updates are computed from. My goal was wall-clock learning speed rather than sample efficiency.

The number of environments and the update ratio were chosen from the sweep in Figure 7, run on the same robot and environment but with the walking task from the CleanRL SAC benchmark [14], and rechecked during reward design.

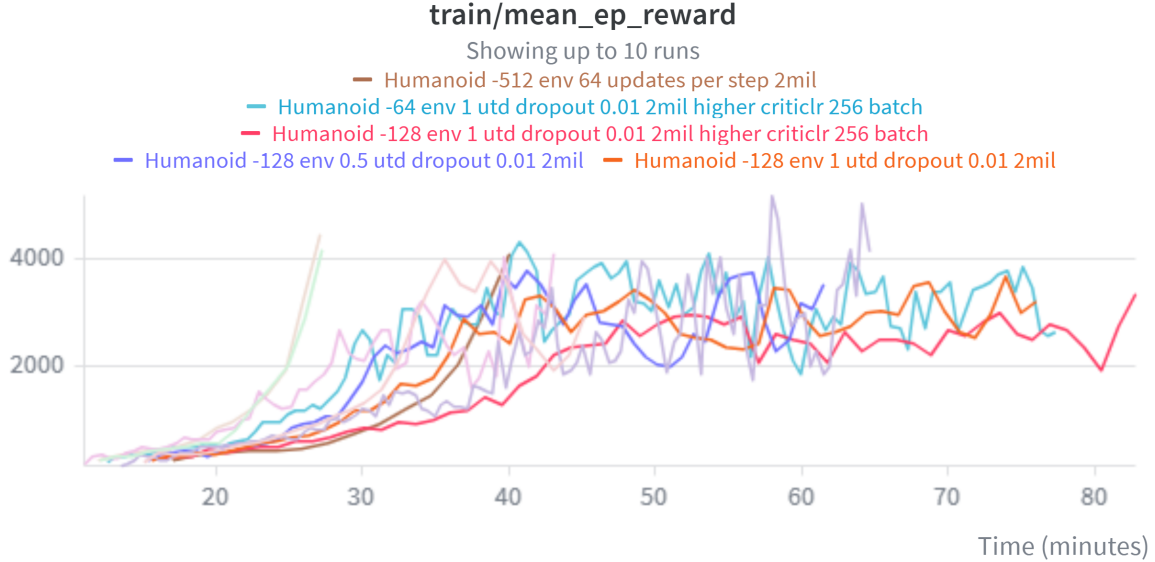


Figure 7: Wall-clock learning speed for different numbers of parallel environments and update-to-data ratios, measured on the walking task [14].

The target-smoothing coefficient is scaled relative to a reference of 16 gradient updates per rollout, so that the target network moves by a comparable amount per unit of collected data regardless of the update ratio. At 128 updates per rollout the nominal value of 5×10^{-3} becomes 6.25×10^{-4} . Twin critics with layer are used to limit overestimation.

During training I logged each reward term separately, so I could see whether the balance the policy realised matched the balance I had specified, and notice when height stopped progressing while some other term kept rising.

5 Reward Design: Failure Modes and Fixes

During experiments different problems and failures arose, and I will explain the most influential changes I had to make, during reward shaping. In Table 7, all experiments are visible.

Table 7: Development series: the change introduced by each experiment and the mean trunk height reached. Spawn height is 1.40 m. The *Base* column names the entry each experiment was derived from. Values marked $\sim$ are the plateau of that phase rather than a logged per-run value; all others are the final logged `mean_trunk_z`.

ID	Addition	Base	Incline	Steps (M)	Mean trunk (m)
E1	First reward	—	0°	20	1.07
E3	Alive bonus removed	E2	0°	20	1.31
E4	Alive bonus restored	E3	0°	20	$\sim$ 1.6
E5	Grip extended to all four limbs	E4	0°	20	$\sim$ 1.6
E6	Per-hold timeout	E5	0°	20	$\sim$ 1.6
E7/1	No alive bonus	E6	0°	20	$\sim$ 1.5
E7/2	Grip observation added	E7/1	0°	20	$\sim$ 1.5
E8	Best height in progress	E7	0°	30	1.24
E9	All hold weights $\times 1.5$	E8	0°	30	1.27
E10	Reverted to E7, scaled up hold weights	E7	0°	30	1.55
E11/1	Positive reach bonus	E10	0°	30	1.57
E11/2	Reach to the top higher holds is higher	E11/1	0°	30	$\sim$ 1.6
E12	Hands height reward	E7	0°	30	$\sim$ 1.6
E13	Leg height reward	E12	0°	30	1.42
E14	Weight sweep (hands, legs, reach)	E13	0°	30	$\sim$ 1.6
E15	Stall ends the episode	E14	0°	30	$\sim$ 1.6
E16	Reverted to the E14 best variant, longer training	E14/3	0°	60	$\sim$ 1.6
E17	Rising lava termination	E16	0°	60	$\sim$ 1.7
E18	4000-step episode	E17	0°	60	$\sim$ 1.6
E20	Soft lava penalty	E17	0°	60	$\sim$ 1.6
E21	More holds	E14/3	0°	60	$\sim$ 1.8
E22	Rectangular holds	E21	0°	30	$\sim$ 1.6
E23	Stickier feet	E22	0°	30	$\sim$ 1.7
E24	Rising lava made terminal	E23	0°	30	$\sim$ 1.8
E25	Reach pays only above the limb	E24	0°	30	$\sim$ 1.5
E26/1 Frameskip 5	E21	0°	60	1.31 ^b	
E26/2	Smaller reach weight	E26/1	0°	60	1.82 ^b
E27	Slow lava	E26/2	0°	60	1.39
E28	More holds	E21	0°	120	2.10
E29	Reach $\rightarrow$ potential-based	E28	0°	60	1.79
E30	Stall truncation	E29	0°	60	1.90
E31	Spawn-height curriculum	E30	0°	60	2.91 ^a
E32	Lifted spawns start attached	E31	0°	60	2.91 ^a
E33	Stronger foot latch	E29	0°	60	1.89
E34	Inclined slab wall [4]	E29	−15°	80	1.58–2.67 ^c
E34'	Same configuration, larger budget	E34	−15°	120	2.27 $\pm$ 0.73 ^d
E36	Per-limb movement schedule	E34	−15°	80	1.84
E42	Stronger hands (gain 200)	E34	−15°	—	—
E43	Frameskip 5, five seeds	E34	−15°	120	3.84 $\pm$ 0.11^e

5.1 Method: Iterative Reward Development

The biggest indicator of success is height progression. I used wandb for logging all the different reward terms separately for clean visibility of the progress of different reward terms, and easier decision making. If there is learning, but the progress reward stalls and others inflate, the experiment needs to change, because of reward hacking. An example of reward hacking is in E14/1, the robot focused on lifting its leg high, which reflected in the leg reward (Figure 8).

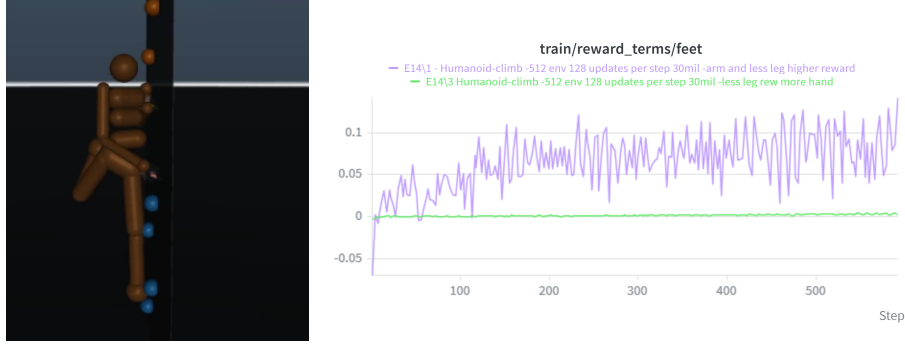


Figure 8: Reward hacking in E14/1. Left: the humanoid holds one leg high without climbing. Right: the feet-height reward stream rises while trunk height stalls.

The experiments were usually run with one seed only, and the purpose was to prove the existence of a behaviour rather than to measure the effect of the change.

5.2 Stationary Reward Streams

After reaching the first hold grabbing and step up, the humanoid learned to stay still. I changed the proximity based to pontential based rewards wherever it was possible, to encourage progression [5]. The reward was for being near the hold, being near the top, rather than approaching them. In E26/1 the reach reward was really high, while in E26/2 I reduced its weight, and observed improvement in torso height (Figure 9).

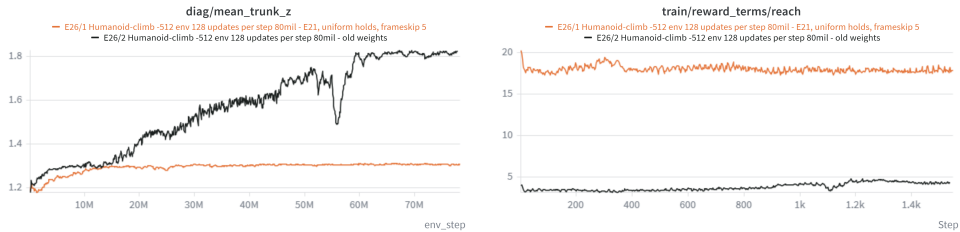


Figure 9: E26/1 against E26/2. Left: mean trunk height. Right: the per-step reach reward, which falls from 16 to 20 to 0 to 5 once its weight is reduced.

The previous reach term fell from a range of 16 to 20 at each logging step in E26/1 to 0 to 5 at E26/2, and the $\text{mean}_t \text{run}_k \text{rose from } 1.3 \text{ to } 1.83$.

The difference is in what each form pays for standing still. Paying Φ directly gives a limb near a hold the same reward at every step, so hanging there is an income stream that never runs out. Paying only the change gives it nothing, because the change is zero.

In E28 I put more hold on the wall, in an attempt to make the task easier. This naturally, increased the reach and the grip reward. E29 differs from its base, E28, in the reach formulation alone. Let

$$\Phi(s) = \frac{1}{|L|} \sum_{\ell \in L} \sum_{h \in H_{\text{active}}} w_h \exp\left(-\frac{d_{\ell h}}{L_{\text{scale}}}\right), \quad L_{\text{scale}} = 0.5, \quad (7)$$

where $d_{\ell h}$ is the surface distance from limb ℓ to the top of hold h and w_h its height weight. Up to and including E28 this quantity was paid directly at every step,

$$r_{\text{reach}}^{\text{old}}(s) = \Phi(s), \quad (8)$$

so the agent was paid for *being* near holds. From E29 onward the same quantity serves as a potential and only its change is paid,

$$r_{\text{reach}}^{\text{new}}(s, s') = \frac{\gamma \Phi(s') - \Phi(s)}{\Delta t}, \quad \gamma = 1, \quad (9)$$

so the agent is paid for *approaching* them. The kernel, the height weights w_h , the length scale L_{scale} and the term weight (1.5) are identical in both; only the quantity paid differs.

I compared the result of 60M steps. The maximum reach contribution fell from 12.09 to 0.60 per step, and mean episode return from $\sim 15\,500$ to ~ 3000 (Figure 10).

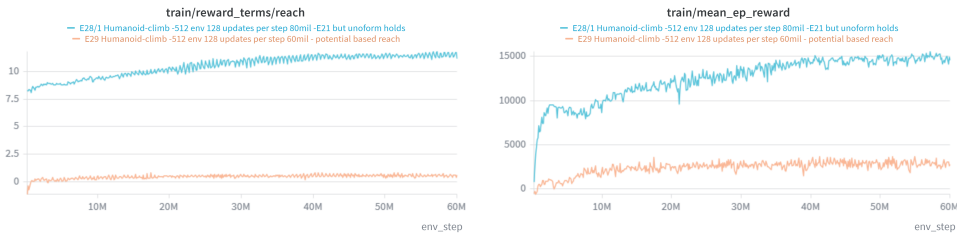


Figure 10: E28 (reach paid directly) against E29 (reach as a potential). Left: per-step reach contribution, 12.09 against 0.60. Right: mean episode return.

E29 was stopped at 60 M environment steps, at which point E28 stood at approximately 1.85 m against E29’s 1.79 m. Over the first 20 M steps E29 led throughout, reaching 1.6 m at 10 M steps where E28 was still at 1.4 m: removing the stream improved early sample efficiency (Figure 11).

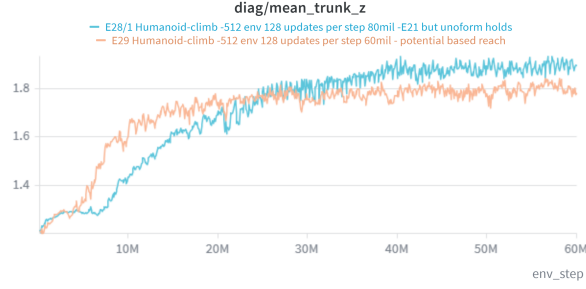


Figure 11: Mean trunk height for E28 and E29. E29 leads over the first 20 M steps and finishes 0.06 m below E28.

Removing more than four fifths of the accumulated return left achieved height approximately the same at equal budget, with no degradation in behavior, both experiments stepped up once. This showed that the removed reward was not paying for climbing.

The same reformulation cannot usefully be applied to the grip term. The problem is with learnability. Grip is a hard indicator, paying w_h when a limb lies within 4 cm of a hold’s surface and nothing otherwise, so the corresponding potential is piecewise constant and its difference vanishes at every step except the one on which a limb crosses that boundary. Where the reach potential is continuous in distance and therefore supplies a gradient at almost every step, a telescoping grip term would supply a single small spike on contact. Grip is therefore bounded in time rather than by telescoping: each hold pays for a fixed period after first contact (Section 5.3), which preserves the magnitude of the signal while still preventing an unbounded stream.

5.3 Hold Camping

Grip and reach were initially defined for the hands alone. E5 extended both to all four limbs, and E6 added a time limit of four seconds after first touch for every hold. All three runs in Figure 12 used the same 20-step budget and are compared only with their own base. E4 reached 1.38 m, E5 1.50 m and E6 1.58 m; the curves separate from roughly forty minutes onward and do not cross again (Figure 12).

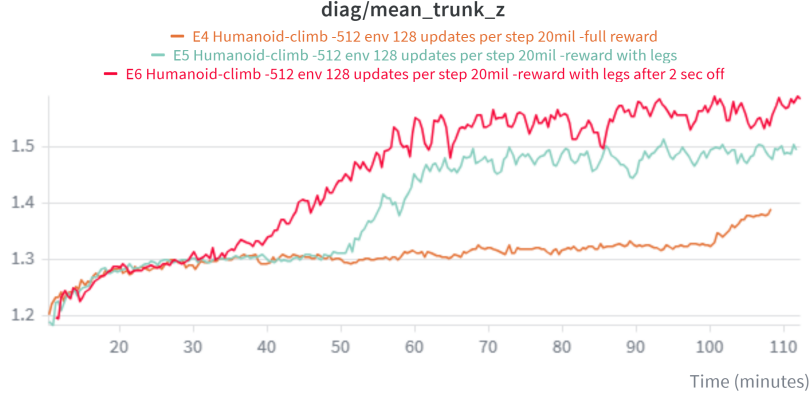


Figure 12: E4, E5 and E6 at a -step budget. Extending grip and reach to all four limbs (E5) and adding the per-hold timeout (E6) each raise final trunk height.

Paying the legs for contact was the larger of the two effects. Limiting the payout was the second.

5.4 Termination types

Penalising stalls to heavily prevents learning, I used episode truncation only [15].

In an attempt to discourage stalling, a rising lava line was introduced as a termination condition at E17 and again at E24, without penalty, on the reasoning that termination removes states a degenerate optimum could occupy [16, 17] and that moving the boundary on a schedule controls which part of the state space is reachable [18]. E27 slowed the lava down. In E30, no stalling was implemented. If the best performance did not improve in 2 seconds the episode is cut short. None of them improves in any way significantly on its base, and no additional step was made by the humanoid. E17 reached 1.7m against E16's 1.8m, E24 1.8m against E23's 1.7m (Figure 13). E27 reached 1.4m against its base E26/2's 1.8m. E30 reached 1.90m against E29's 1.79m (Figure 14).

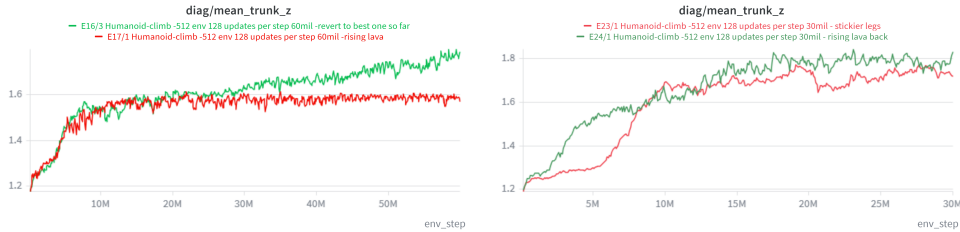


Figure 13: Rising-lava termination. Left: E17 against its base E16. Right: E24 against its base E23. Neither produces a further step up.

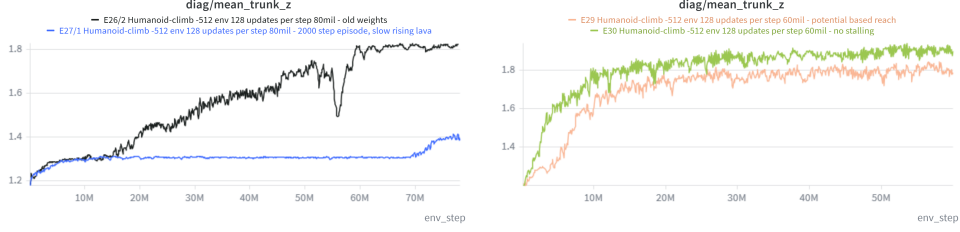


Figure 14: Left: E27, slowed lava, against its base E26/2. Right: E30, stall truncation, against its base E29.

Table 8 collects the four interventions against their own bases.

Table 8: Interventions on episode structure and termination, each compared with its own base.

Run	Change	Base (m)	Result (m)
E17	Rising lava as termination condition	1.6	1.7
E24	Rising lava as termination condition, no penalty	1.7	1.8
E27	Slow rising lava	1.8	1.4
E30	Stall truncation	1.79	1.90

Taken together these rule out an explanation rather than supply one. If the humanoid stopped climbing because remaining still was tolerable, then making it intolerable by ending the episode, by killing the agent from below, or by charging a penalty for falling behind a rising line, should have produced movement. Four variants did not therefore I moved the search to the reward composition (Section 5.2) and to the environment itself.

5.5 Lowest-Hold Bias

The robot consistently grabbed the closest and lowest holds, because a flat grip and reach bonus pays the same for each hold. The cheapest hold to reach is also the most valuable one, and nothing in the reward distinguishes a hold that advances the climb from one that does not. To reach for holds that enable climbing, in E11/2 reach was scaled by a hold-height weight w_h , running from 1.0 at the lowest hold on the route to 2.5 at the highest, a gain of 1.5 above a base of 1.0), so that approaching or latching a hold near the top of the wall pays up to two and a half times what the same action pays at the bottom. The humanoid still grabbed the lower holds, and no significant improvement was visible in height in E11/2. In E12 I introduced an arm height reward, which did not change height, and did not achieve a next step, but the the desired behaviour was observed, the humanoid touched higher holds (Figure 15).



Figure 15: Left: E11/2, with height-weighted grip and reach; the humanoid still takes the lowest holds. Right: E12, with the hands-height term; it touches higher holds without gaining trunk height.

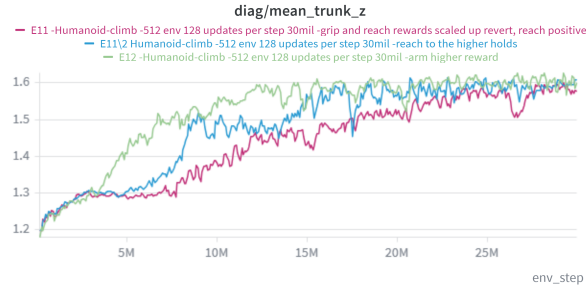


Figure 16: Mean trunk height for E10, E11/1, E11/2 and E12.

E11 reached 1.57 m against E10’s 1.55 m, E11/2 approximately 1.6 m against E11’s 1.57 m, and E12, which added a hands-height term, approximately 1.6 m against E7’s 1.5 m (Figure 16).

5.6 Weights

In E12 I added an arm height reward to encourage reaching higher, In E13 a leg height reward, to encourage stepping on higher holds further. E13 reached 1.42 m against E12’s approximately 1.6 m, but I run more trainings to see what weights would work. In E14/1 -5 different weight variations were observed for the hand leg and reach rewards (Table 9). The torso height was roughly the same, except E14/5, which struggled to find the first step (Figure 18). With too big leg weight, the humanoid stepped up once, and lifted one of its legs high in E14/1. During E14/3 a behaviour was observed, where the robot steps up two times (Figure 17).



Figure 17: E14/3, the first configuration in the series to step up twice.

The goal was to find proof of existence to a behaviour, not measuring.

Table 9: Weight variants evaluated at E14, all on the E13 base at seed 53. The feet term is the one described as the leg reward. Variants 1–3 ran to 30 M steps and variants 4–5 to 60 M; variant 4 repeats variant 3’s weights at the larger budget. Weights not listed are identical throughout: grip 0.5, progress 10, torque 0.02, smooth 0.01.

Variant	hands	feet	reach	Trunk (m)
E14/1	3	1.00	1.0	1.579
E14/2	3	0.25	1.5	1.646
E14/3	4	0.05	1.5	1.694
E14/4	4	0.05	1.5	1.550
E14/5	5	0.05	1.5	1.735

The E14/3 weight for legs is 0.05 against 4 for hands and 1.5 for reach, nearly switching off the term E13 had introduced. These weights were carried forward and are unchanged in every subsequent run reported in this thesis, including E43, the most successful setup. The selection rests on a behaviour observed in two of the four. runs of this configuration.

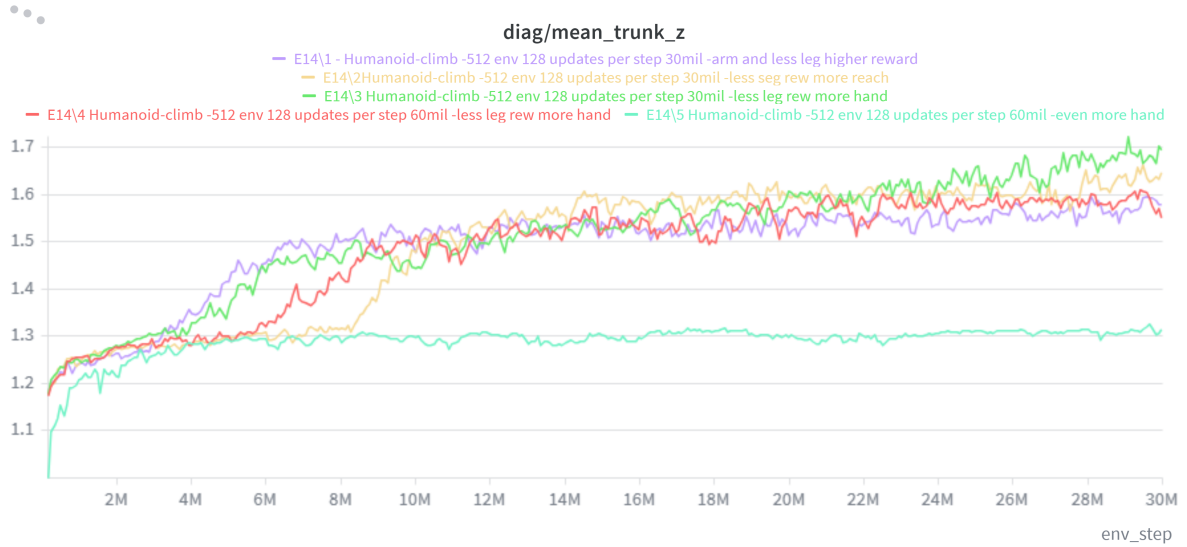


Figure 18: Mean trunk height for the five E14 weight variants. All but E14/5 converge to roughly the same height.

In E16 I trained E14/3 with 60M steps, and different seeds. E16 re-ran the E14/3 configuration three times at 60 steps. Only 1 out of three, E16/3 stepped up. The three runs reached 1.595, 1.604 and 1.789 m (Figure 19).

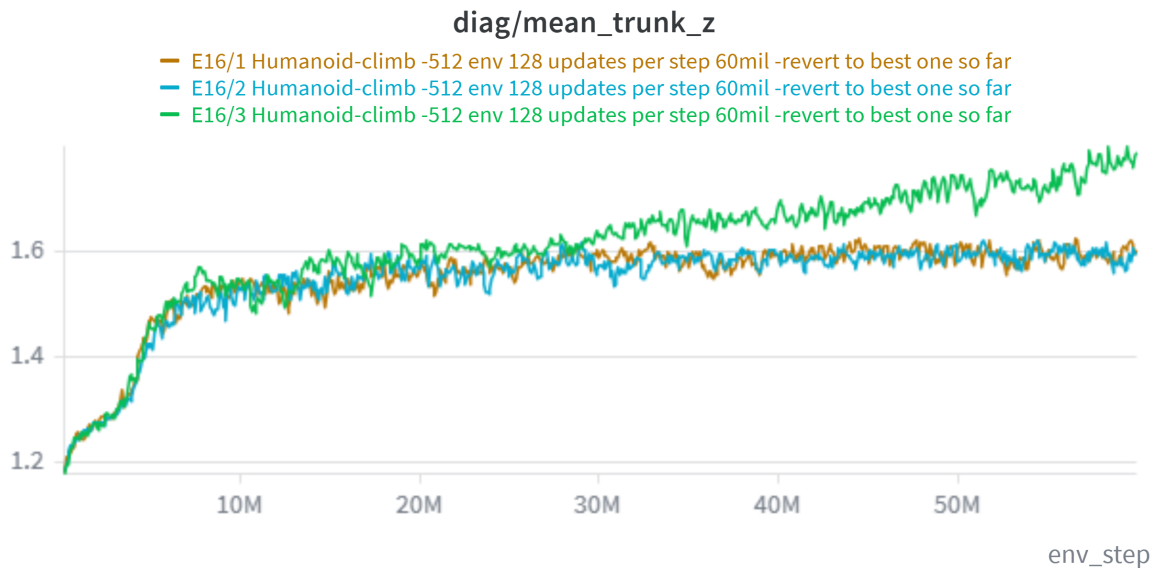


Figure 19: E16: three re-runs of the E14/3 configuration at 60 M steps. Only E16/3 steps up.

5.7 Environment changes

Wall geometry In E22 I replaced the spheres with rectangular holds, but the torso height degraded from 1.8 to 1.6, and while its base, E21 stepped up twice in 30M steps, E22 only stepped up once, therefore I decided to switch back to soherical holds (Figure 20).

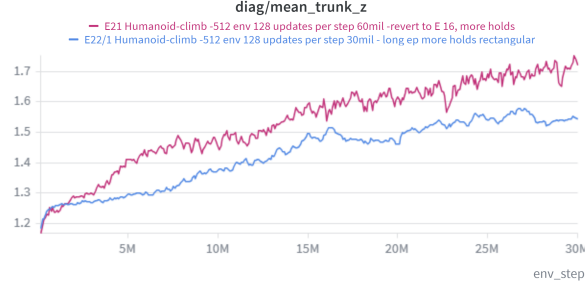


Figure 20: E22, rectangular holds, against its base E21, spherical holds. Trunk height degrades from 1.8 to 1.6 m.

All experiments before E34 used a vertical wall with a hand-placed route. In E34 I replaced it with an inclined slab based on Kang et al. [4], a -15° inclined wall with a uniform 12×4 grid of spherical holds at 0.4 m spacing on it. E34/2 was the first run that achieved continous climbing, getting the torso to 3.2 against E29’s 1.79 m in 60M steps (Figure 21). Every result above 2.2m in this thesis was obtained on the slab.

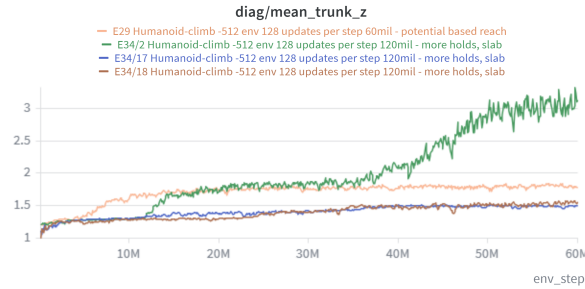


Figure 21: E34/2 on the inclined slab against E29 on the vertical wall. E34/2 is the first run to climb continuously.

Frameskip The climbing task was initially given a frameskip of 2, and therefore 0.006 s, on the reasoning that climbing is contact-rich in a way locomotion is not, a limb must be placed within 4 cm of a hold’s surface for the grip to engage, contact events are brief, and a fall terminates the episode. A finer control interval gives the policy more opportunities to correct within a move and reduces the chance of stepping over a contact transition. Frameskip was changed to 5 at E43, on the slab, for two reasons. It is the standard value for this humanoid, and at a fixed cap of 1000

steps it affords 15 s of simulated time per episode against 6 s. Lengthening the episode to 15 s, by simply increasing an episode steps, would produce a substantially greater RL overhead for the lower-frameskip configuration. Both configurations require 5000 physics substeps to simulate the same 15 s, but at frameskip 2 this corresponds to a 2500-step episode and 1000 steps at frameskip 5. Because rollouts contain a fixed number of environment steps, the two configurations perform the same number of policy evaluations, store the same number of transitions, and perform the same number of gradient updates at a fixed environment-step budget. This means frameskip 5 provides 2.5 times as much simulated time for the same training budget. Table 10 sets out what the longer control interval does and does not change.

Table 10: What changes and what does not when the control interval is lengthened. Gradient updates are performed once per rollout, and a rollout is a fixed number of environment steps (16 steps across 512 environments), so the compute spent is set by the environment-step budget and is identical in both columns. What differs is how much simulated experience each environment step buys. Figures in the last group assume the 120 M-step budget and 1000-step episode cap used in E34 and E43.

	Frameskip 2	Frameskip 5
<i>Identical: compute per rollout</i>		
Environment steps per rollout (all envs)	8192	8192
Gradient updates per rollout	128	128
<i>Differs: simulation per environment step</i>		
Physics substeps per environment step	2	5
Control interval Δt	0.006 s	0.015 s
Simulated time per rollout, per environment	0.096 s	0.24 s
<i>Consequences</i>		
Episode duration at a 1000-step cap	6 s	15 s
Discount horizon, 100 steps at $\gamma = 0.99$	0.6 s	1.5 s
Total gradient updates at 120 M steps	1 875 000	1 875 000
Total simulated experience at 120 M steps	720 000 s	1 800 000 s

The discount horizon is measured in steps rather than in time. At $\gamma = 0.99$ the effective horizon is approximately 100 steps, which is 0.6 s at $\Delta t = 0.006$ s and 1.5 s at 0.015 s. A single climbing move takes on the order of a second, so at frameskip 2 rewards for any completed move are more heavily discounted over the duration of the move than at frameskip 5. Increasing the episodes length with more steps does not change this step-based discounting, increasing the control interval does.

The change of the control interval changed the reliability of the outcome. In E34, three of ten runs at frameskip 2 achieved continuous climbing, stepping up 5+ times, reaching 2.67, 3.23 and 3.37 m, while the remaining seven converged between 1.58 and 1.91 m.

Figure 22 shows five of E34 experiments, which were completed in 120M steps.

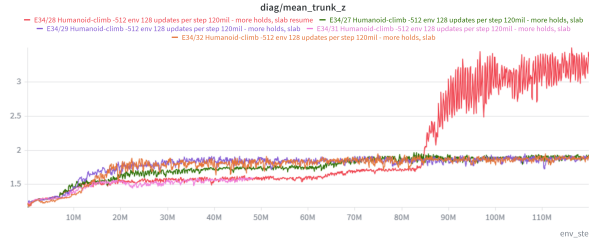


Figure 22: Five E34 runs at frameskip 2, each to 120 M steps.

All five runs at frameskip 5 reached 3.71 m to 3.98 m, and the humanoid touched the top row of holds in all of them. The reward functions and other environment settings are the same for all fifteen runs. The highest frameskip-2 result is below the lowest at frameskip 5, and the mean of successful runs rises from 2.16 to 3.84 m.

In experiments with frameskip 2, it was common for the humanoid to complete a move and hold position, which could suggest that the episode length was the problem. I did one training, where only the episode length differed. In E18 I raise the maximum environment step per episode to 4000 from 1000 in E17, but both of them had rising lava installed, which never allowed episodes in E18 to be longer than 850 steps.

Separating the effects, requires more experiments holding simulated time fixed while varying the control interval, which this series does not do.

5.8 Specified Weights and Realised Reward

Throughout the series I logged each reward term separately. Looking back over the runs where the terms were logged, one comparison is worth making, and one is not.

The one that is not is between a term and the height reached. Progress, hands and feet are telescoping height potentials, so what each pays over an episode is the rise of the torso, of the hands and of the feet. A run that climbs higher collects more of all three by construction. Correlating them with trunk height says nothing that the definitions do not already say. Only grip and reach are independent of height, and both are weighted by hold height, so even they are not fully separate from it.

What can be compared is the reward as I specified it against the reward the policy actually collected. E34 gives ten runs with identical weights, on the same wall, at the same control interval, differing only in seed. Three of them climbed and seven did not. Table 11 gives the share each term contributed to the total positive reward per step.

Table 11: Share of positive per-step reward contributed by each term, for the ten E34 runs at frameskip 2 and the five E43 runs at frameskip 5. All runs use identical weights: progress 10, hands 4, reach 1.5, grip 0.5, feet 0.05.

Trunk (m)	progress	hands	feet	reach	grip
<i>E34, frameskip 2</i>					
3.37	40.1%	16.2%	0.2%	16.3%	27.1%
3.23	43.4%	15.3%	0.2%	16.9%	24.1%
2.67	41.3%	15.9%	0.2%	19.2%	23.3%
1.91	16.5%	13.2%	0.1%	26.1%	44.0%
1.91	15.9%	11.5%	0.1%	26.0%	46.5%
1.89	15.9%	12.4%	0.1%	24.9%	46.6%
1.85	14.4%	7.4%	0.1%	27.1%	50.9%
1.60	6.0%	11.2%	0.1%	27.7%	55.0%
1.59	5.6%	6.1%	0.1%	34.6%	53.6%
1.58	5.8%	6.1%	0.1%	25.8%	62.2%
<i>E43, frameskip 5</i>					
3.98	37.6%	16.0%	0.2%	15.2%	31.0%
3.89	34.8%	15.1%	0.2%	17.2%	32.7%
3.86	34.3%	14.1%	0.2%	17.9%	33.6%
3.76	33.0%	13.6%	0.2%	16.2%	37.0%
3.71	35.5%	14.8%	0.2%	14.8%	34.7%

The weights say progress should be worth twenty times grip: 10 against 0.5. In the three E34 runs that climbed, progress delivered 40 % to 43 % of the positive reward and grip 23 % to 27 %, so progress paid roughly twice grip. In the seven that did not, progress delivered 6 % to 17 % and grip 44 % to 62 %, so grip paid between three and ten times progress. The ratio I specified is 20 : 1. The ratio the plateaued runs realised is closer to between 1 : 3 and 1 : 10.

Nothing in the reward differs between those ten runs. What differs is which balance the policy settled into. A weight sets what a behaviour is worth if the policy performs it, not what the policy ends up collecting, and a policy that stops climbing collects almost nothing from the term that was supposed to dominate. Grip, the smallest weight in the reward apart from the effort terms and the feet, became the largest realised contribution in every run that failed.

The five E43 runs make the same point from the other side. Their shares are almost identical to one another, between 33 % to 38 % progress and 31 % to 37 % grip, so the composition is consistent across seeds, where in E34 it was not. Once the control interval was lengthened, every seed converged to the same balance, and it was a balance in which the task term paid more than the contact term.

This is why the per-term logging was the instrument I relied on rather than the return. Two runs with the same reward, the same wall and the same budget can differ by a factor of two in height, and the reward is spent on entirely different things in each. That is not visible in a single number.

6 Results

6.1 Experimental Setup

I used the simulation settings described in Section 3, and the reward described in Section 4.3. I ran 5 independent training runs with different seeds. Each was trained for the full budget of 1.2×10^8 environment steps and all five completed it. They all reached the top row of holds, with their hands. GPU execution is not bit-deterministic, so identical seeds do not reproduce identical runs.

6.2 Evaluation Metrics

The main evaluation metric is torso height, as that can only be meaningfully increased via climbing. Torso height is reported as the mean over the 512 parallel environments at each logging step. The reward terms are designed and tuned to encourage the desired behaviour, but they can also be exploited, as discussed in Chapter 5. For evaluation, height is therefore a more meaningful metric. Height is supplemented by two secondary measures. Episode length indicates how often an episode ends in a fall. However not fall could also mean stalling, so it is rather an indicator of frequency of failure, not a performance measure. Highest hold reached shows if the robot truly got to the top.

6.3 Climbing Performance

The trainings lead to the robot reaching from 1.4 spawning height to 3.84 ± 0.11 meters via climbing with 3.98 m highest and 3.71 m lowest. The means of returns, torso heights and episode lengths achieved during the last episode of the training, are visible in Table 12. The data was collected during stochastic exploration.

Table 12: Final performance of the five runs. Spawn height is 1.40 m; episodes are capped at 1000 steps.

Seed	Mean trunk height (m)	Mean episode reward	Mean episode length
5	3.86	4900	954
6	3.89	5321	966
7	3.76	5127	982
10	3.98	4365	938
12	3.71	4947	987
Mean $\pm$ sd	3.84 ± 0.11	4932 ± 358	965 ± 20

In Figure 23 we can see that the robot learns to survive almost the full 1000-step episode relatively early in the training, and maintains long episodes after. Survival is acquired well before

climbing, and episode length is not a good indicator of progress, standing still can survive as well. Two runs had to be cancelled for technical reasons, therefore the names skip E43/4 and E43/5.

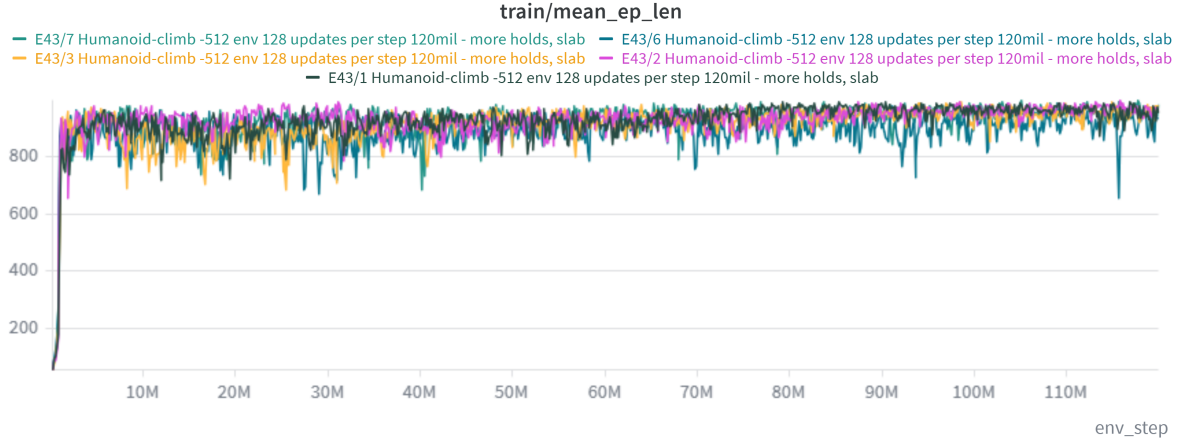


Figure 23: Mean episode length over wall-clock time for the five E43 runs. Survival is acquired well before climbing.

If Figure 24 a plot of the torso’s height is depicted over wall clock time. The results show a lot of variation in the learning dynamics between runs, despite the similar final performance. The policies improve in multiple separate stages, with periods of rapid progress followed by plateaus.

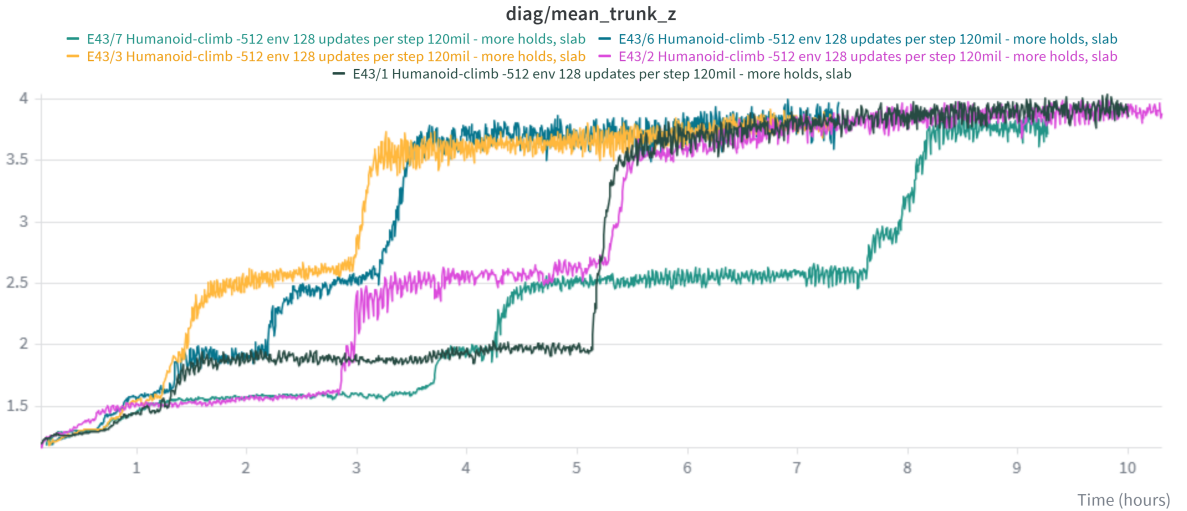


Figure 24: Mean trunk height over wall-clock time for the five E43 runs. Learning proceeds in stages, with plateaus between periods of rapid progress.

In Figure 25 we can see the mean episode return over wall clock time. This show more fluctuation, than Figure 24, however there is correlation in the plateaus and progression points. For

example all runs start rapid progression around 1 hour for in both plots.

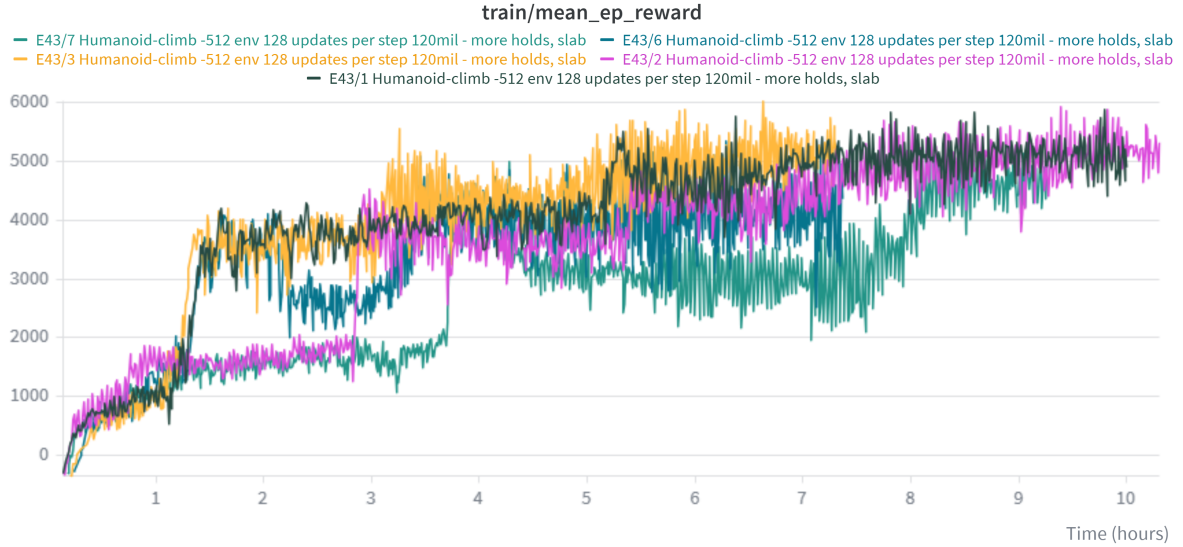


Figure 25: Mean episode return over wall-clock time for the five E43 runs. Noisier than Figure 24, but the plateaus and progression points coincide.

The videos of the final actors evaluation episodes are available at Appendix A. The climbing is leg driven in all five runs, and this is not explained by the arms being too weak. At 80 N m a two-arm pull is inside the actuator envelope with room to spare. The policy had the option and did not take it. Why it did not is not established here: the hip actuators are stronger still at 120 N m, and the reward does not distinguish between limbs, so either could account for it.

For seed 5 the robot gets to the top in 8-10 seconds, stepping up slowly. It steps up one row higher than the other. It lifts its left leg very high every step. There is some jittering at the end, and the robot pulls itself really close to the wall (Figure 26).

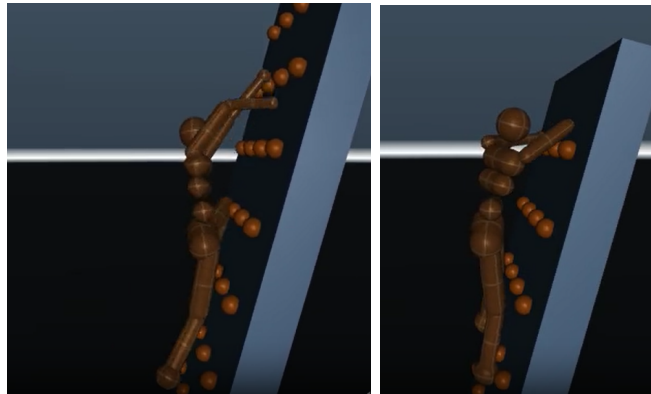


Figure 26: Seed 5. Left: the left leg raised high during a step. Right: at the top of the route, one row above the other seeds.

For seed 6 the humanoid gets to the top in 7-8 seconds (Figure 27). Steps up with the left leg first, but also uses right. There is jittering at the top, but at the end it pulls itself inwards.

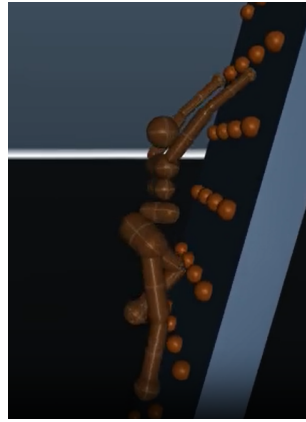


Figure 27: Seed 6, mid-route.

The humanoid steps up with the left leg first, right leg is also used. There is jittering at the top, but at the end it pulls itself inwards. (Figure 28).

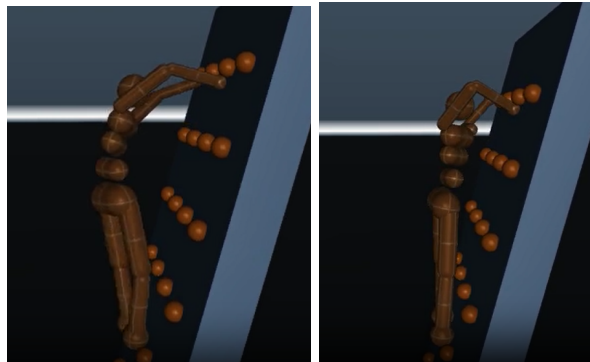


Figure 28: Seed 6 pulling itself in towards the wall. Left: before. Right: after.

For seed 7 the robot gets to the top in 4-5 second. It uses the hips a lot, and step up more using the right leg than the left. At the top there is some jittering, when it tries to adjust position during slipping (Figure 29).

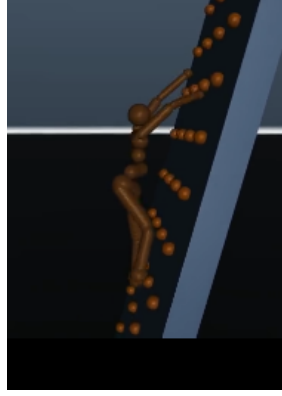


Figure 29: Seed 7, which uses the hips heavily and leads with the right leg.

For seed 10, the humanoid climbs to the top in about 5-6 seconds. There is repetition when stepping to new holds (Figure 30). It always uses the left leg, stepping high. After reaching the last hold the humanoid keeps hanging on the top holds, sometimes changing position to prevent slipping, touching new holds in the process.



Figure 30: Seed 10 stepping to a new hold.

For seed 12 mostly the left leg is used. The hands are iterating. The humanoid reaches the top in 7-8 seconds. There is jittering and sliding at the top (Figure 31).

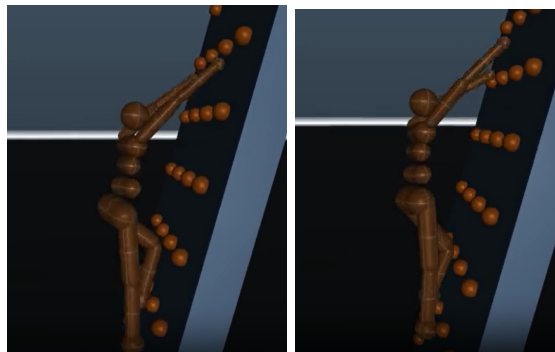


Figure 31: Seed 12, which leads with the left leg and alternates hands.

6.4 The Role of the Grip Assist

The grip assist needs to be strong enough to help the body hang for a few seconds, but cannot be unescapable. The magnetic assist can look very unnatural. It can happen that the feet is not on top of the hold, but under, but it still sticks to the hold, making it look like the leg is hovering on the wall seemingly without anything holding it. This can be seen in Figure 32.

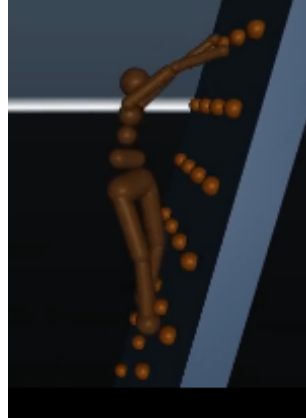


Figure 32: A foot latched to the underside of a hold. The assist holds it there, so the leg appears to hover against the wall.

6.5 Evaluation on Route Variance

The training curves say the policies climb the route. They do not say what the policies learned in order to do it. A policy that picks a hold and moves to it, and a policy that repeats one fixed step until it runs out of wall, produce the same curve on a uniform grid.

The evaluation videos point at the second. The same leg leads almost every move and the same step repeats instead of a varied sequence. This section tests that directly. If the policy is choosing moves, it should tolerate any change that leaves the holds reachable. If it is repeating one stride, it should tolerate whatever leaves that stride valid and fail on everything else. The two predictions differ, so the experiment can separate them.

I took two of the trained actors and ran them on walls they had never seen. Each wall was evaluated for ten episodes with the actor acting deterministically, so the only variation between episodes is the reset noise on joint positions and velocities. I varied one property at a time: the inclination of the wall, its distance from the spawn point, the vertical and horizontal spacing of the holds, the number of columns, the radius of the holds and how far they stick out, random jitter on every hold position, and removing a fraction of the holds. E43/7 was used for most of the walls and E43/10 for the jitter and offset walls, so the two checkpoints are reported separately and are not compared with each other. The different types of wall are visible in Figure X.

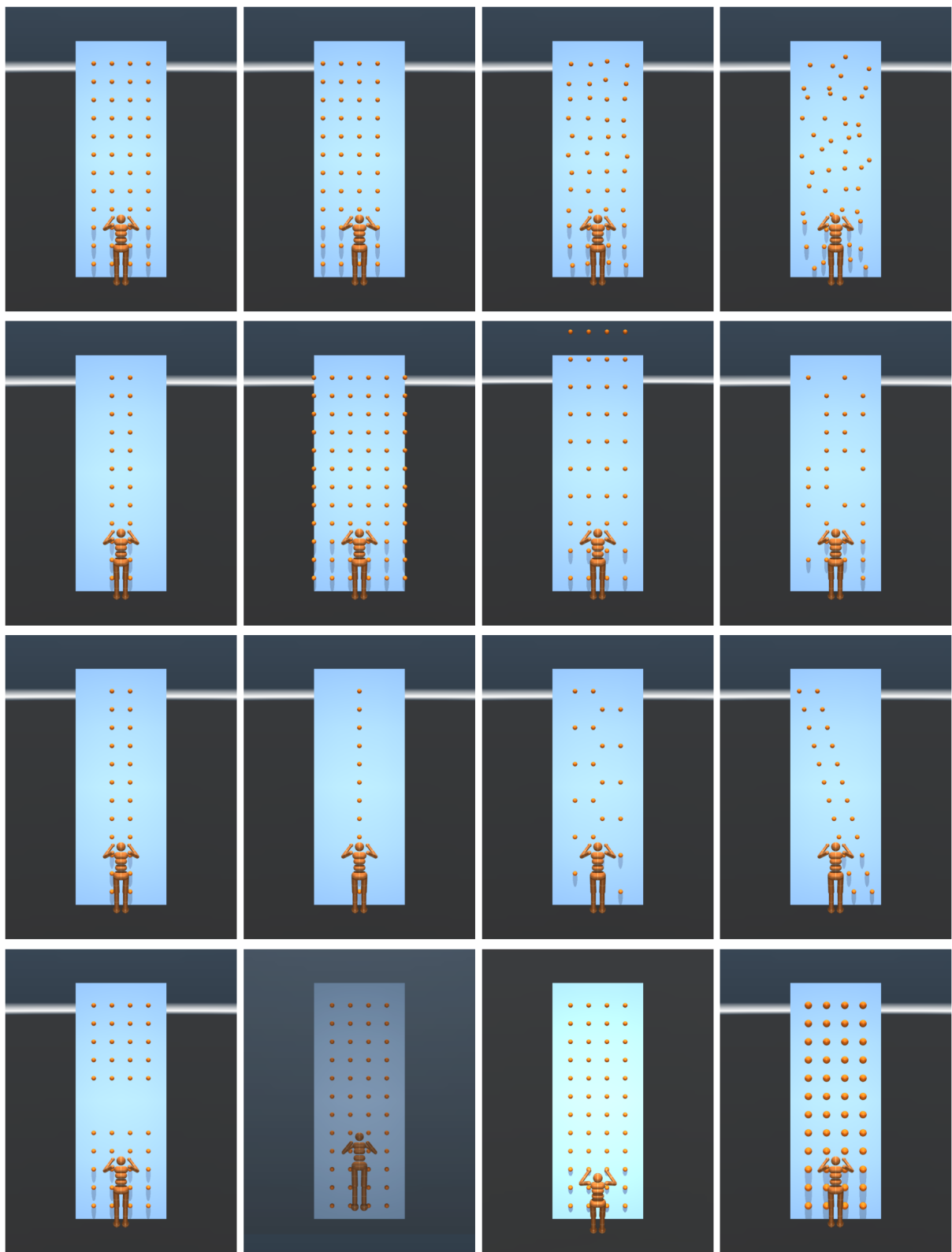


Figure 33: Different walls used

The baseline On the training route E43/7 gained (2.883 ± 0.021) m over ten episodes and reached the top row in all ten. E43/10 gained (2.909 ± 0.010) m. A spread of two centimetres across ten episodes means the policy is close to deterministic on the wall it was trained on, and that any drop on another wall is an effect rather than noise.

Table 13: Trained policies evaluated on modified routes, ten episodes each, deterministic actor. *Gain* is the mean rise of the torso above its spawn height. *Top* counts the episodes gaining at least 2.5 m, which on this route means reaching the top rows. Each block is compared only with the baseline above it.

Route	Change	Gain (m)	vs base	Top
<i>E43/7</i>				
kang	training route	2.883 ± 0.021	—	10/10
x45	wall 5 cm nearer	2.884 ± 0.025	+0%	10/10
cols6	6 columns instead of 4	2.612 ± 0.517	−9%	9/10
cols2	2 columns instead of 4	2.568 ± 0.495	−11%	9/10
prot05	holds come out of wall further	2.577 ± 0.513	−11%	9/10
x55	wall 5 cm further	2.360 ± 0.649	−18%	8/10
dy30	columns 0.30 m apart	1.896 ± 0.421	−34%	1/10
prot01	holds nearly flush	1.519 ± 0.692	−47%	3/10
r08	hold radius 0.08 m	0.939 ± 0.217	−67%	0/10
dv30	rows 0.30 m apart	0.866 ± 0.314	−70%	0/10
r03	hold radius 0.03 m	0.348 ± 0.195	−88%	0/10
dv50	rows 0.50 m apart	0.258 ± 0.008	−91%	0/10
dy50	columns 0.50 m apart	0.163 ± 0.128	−94%	0/10
drop20	20% of holds removed	0.169 ± 0.022	−94%	0/10
dv60	rows 0.60 m apart	0.103 ± 0.038	−96%	0/10
drop40	40% of holds removed	0.000 ± 0.000	−100%	0/10
jitter10*	holds jittered ± 10 cm	$0.004\text{--}0.442$	−100% to −85%	0/10
<i>E43/10, 400-step cap</i>				
kang	training route	2.908 ± 0.006	—	10/10
jitter05*	holds jittered ± 5 cm	$1.160\text{--}2.425$	−60% to −17%	0–6/10
shift20	route shifted laterally	0.043 ± 0.038	−99%	0/10
phase20	rows offset in phase	0.021 ± 0.037	−99%	0/10

*Range over five jitter seeds, each evaluated for ten episodes, together with their mirrored variants for jitter10.

What survives. Five changes cost almost nothing. Moving the wall five centimetres nearer had no effect at all. Adding two columns cost 9 % and removing two cost 11 %, and in both cases the policy still reached the top in nine episodes out of ten. Making the holds stick out further cost 11 %. Moving the wall five centimetres away cost 18 % and reached the top eight times.

All five leave the vertical structure of the route alone. The rows stay at the same heights and the same spacing.

What breaks. Everything that touches that structure. Bringing the rows from 0.40 to 0.30 m cost 70 % and no episode reached the top. Spreading them to 0.50 cost 91 % and to 0.60 cost 96 %. Removing a fifth of the holds at random cost 94 %, and removing two fifths stopped the humanoid moving at all. Jittering every hold by up to ten centimetres, which leaves the grid

roughly where it was, reduced all ten jitter walls to under half a metre of gain.

The jitter walls give a rough dose response. At ± 5 cm the policy still climbed some of the time, between 1.16 and 2.43 m depending on the jitter seed, with up to six episodes in ten reaching the top. At ± 10 cm nothing reached the top on any of the ten walls. The two sets used different checkpoints and different episode caps, so this is indicative rather than controlled, but the direction is clear.

Inclination. The policies were trained on a -15° slab. At -20° and at -10° the gain fell by more than 90 % for both checkpoints, and at -30° the humanoid did not leave the ground. Five degrees in either direction removes the behaviour.

What this says the policy learned. The second prediction is the one the data supports. The policy is keyed to the vertical pitch of the holds, and to almost nothing else. Changing it by ten centimetres the climbing behaviour cannot correct. Changing how many columns there are, or how far away the wall is, leaves it untouched, because a policy that steps up through rows never had to choose between columns in the first place.

That is what a fixed stride looks like from the outside, and it agrees with the videos: one step repeated at one height, led by the same leg, rather than a move selected for the holds in front of it. It also explains why removing holds hurts so much more than moving the wall. A missing hold breaks the stride at one row, and there is no second choice to fall back on.

This means the contact sequence was discovered from the reward and not supplied. But what was discovered is one sequence for very similar walls only. Generalisation was never a goal here, and the training route was deliberately uniform to make the task learnable. The cost of that choice is what this table measures.

7 Discussion

7.1 Answers to the Research Questions

Can humanoid climbing be learned from a shaped reward alone, in a single stage, without a planner, a pose dataset or a motion prior? In E43 all five runs climbed the route, taking the torso from its spawn height of 1.40 m to (3.84 ± 0.11) m, and in every run the humanoid reached the top row of holds. No stance planner, no hanging-pose dataset, no captured motion to imitate, and no target position were supplied, but the policy struggles on unseen walls.

Which reward formulations produce climbing, and what goes wrong? Chapter 5 lists the failure modes I ran into: a proximity term that paid for staying near a hold instead of approaching one, holds that kept paying after they had been used, a flat bonus that made the lowest hold as valuable as the highest, and a leg-height term that paid for lifting a leg without climbing. Each has a fix and each fix is described there. A lot of the rewards configurations produced one single step, but only the final reward achieved continuous climbing to the top row.

How reliable are single runs? Not very. In E43, five different seeds gave a standard deviation of 0.11 m. Most of the differences between neighbouring experiments in Table 7 are smaller than that. Only two comparisons in the whole series are larger than the spread and have more than one run on each side: the change of wall and the change of control interval.

7.2 What the Extra Structure Buys

The previous work gives the agent more than I did, and climbing still happened without it. What the extra structure buys is something else.

Kang et al. [4] initialise their climbing policy from a hanging policy and shape its style with a discriminator trained on captured human climbing. Naderi et al. [2, 3] supply the contact sequence. My results suggest that none of this is needed to get the character up the wall. What it supplies instead is the quality of the motion, which is what those papers set out to produce.

The grip, however, is abstracted. The humanoid has no hands and no fingers, and a hold is taken by a distance-triggered attractor that pulls the limb onto it and keeps it there. The agent does not learn to grasp, and once a limb is within 4 cm of a hold it cannot fail to grasp. Climbing has two problems, where to place the limbs and how to hold on, and this removes the second of them. Some form of this simplification is made throughout the literature. Two differences from that prior work are worth noting. In Kang et al. [4] the policy emits a grip signal and releases by deactivating the joint, whereas here the attractor engages automatically whenever a limb comes within 4 cm, so gripping is not an action the agent takes. And the attachment is compliant rather than rigid: a limb can be pulled free (Section 3.4), and a ball joint cannot be.

The climbing my policies produce works and does not look like climbing. The evaluation videos show the same leg leading almost every move, the same step repeated rather than a varied sequence, jittering at the top of the route, and the body pulling itself flat against the wall. Kang et al’s exists to remove these unrealistic movements, and their reported motion is more natural and more varied than mine. The scaffolding in the prior work is therefore not what makes the task solvable, but using it helps to achieve more natural movement and even limb usage.

7.3 Limitations

The episode is short. An episode is 1000 steps, which at $\Delta t = 0.015$ s is 15 s of simulated time. The fastest of the final runs reaches the top in about four seconds and the slowest in about ten, so the cap is not binding for the final configuration.

One wall, one morphology. A single route family was used during training. The aim was to establish whether the task is learnable without extra support, not to produce a route-general controller. All results use the Gym humanoid on a -15° slab with a uniform grid of identical spherical holds

Part of the observation is dead. The observation reserves 84 of its 392 dimensions for external contact forces, taken from `cfrc_ext`. MJX does not populate that field, so those 84

values are identically zero in every run reported here. I found this while analysing the results, after the successful trainings had already been run. The task is still solvable because the grip depends on distance to the hold rather than on contact force, and the grip indicator bits report attachment directly. But the policy had no contact-force information at all, and more than a fifth of the observation carries nothing.

Because simulation and learning updates execute on GPU with non-deterministic reduction order, runs sharing a seed would not reproduce one another exactly.

7.4 Threats to Validity

The development series is not a controlled ablation. Many entries use different step budgets and most comparison has a single run on each side. The measured spread is 0.19 m between identical runs. Most differences between neighbouring experiments in this thesis are smaller than that. A single run therefore cannot tell you whether a difference came from the change or from the randomness. The two comparisons that survive this are the change of the wall at E34 and the change of control interval at E43.

8 Conclusion

This thesis asked whether a humanoid can learn to climb without a stance planner, a bootstrapped pose dataset or a motion prior trained on captured climbing in one training stage, with the contact sequence discovered from the reward rather than supplied. In the final configuration all five runs climbed the route, and in every one the humanoid reached the top row of holds, though the grip itself is abstracted rather than simulated, and only very similar walls can be climbed.

It took forty-three experiments to find. Thirty of them, spanning reward reformulations, five termination types, different hold and wall geometries produced an agent that, on its final episode rose the mean height of the humanoid across 512 environments to halfway on a vertical wall. Multi-move climbing appeared during that phase, but continuous climbing did not.

Replacing the wall with an inclined slab and lengthening the control interval each moved the outcome. The latter, isolates one factor with several runs on each side: three of ten runs climbed at a frameskip of 2 and five of five at a frameskip of 5.

In the final configuration the humanoid spawns on the ground in front of the wall and finishes with its hands on the top row of holds, having been given no route, no poses to reproduce and no sequence of contacts to follow.

8.1 Future Work

Separate the control interval from the episode duration. The largest effect in this thesis conflates two things. Running frameskip 2 with a 2500-step cap gives the same 15 s of simulated time per episode as frameskip 5 with 1000 steps, and comparing the two would say which of the pair matters. Three seeds would be enough. This is the cheapest remaining experiment and it

addresses the thesis’s main result.

Ablate the reward terms. No experiment here removes a single reward term and holds everything else fixed. Doing so for the feet, hands, reach and grip terms, at three seeds each against the E43 baseline, would answer directly what each contributes. The route is saturated at 3.84 m, so time to the top and number of holds contacted would be more informative than height.

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A Videos

E43 result videos can be found here:

<https://drive.google.com/drive/folders/1WXYuPqIM0agqsYsJTZvb7onToY8tobi8?usp=sharing>

B Source code

The simulation environment, the training pipeline and the evaluation scripts are available at

https://github.com/leakaktusz/humanoid_climbing.

C Tools used

I used Claude Code during the thesis to assist with debugging and resolving errors in code, and revising sections of the written text. I used ChatGPT to revise written text.